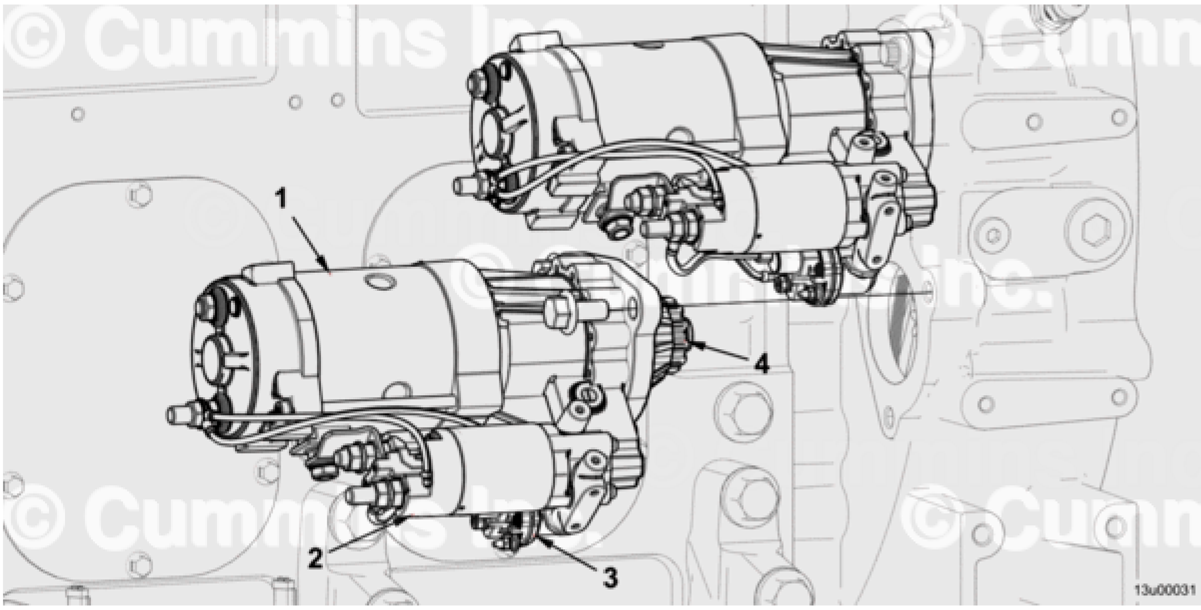
Trainings

Exploded View

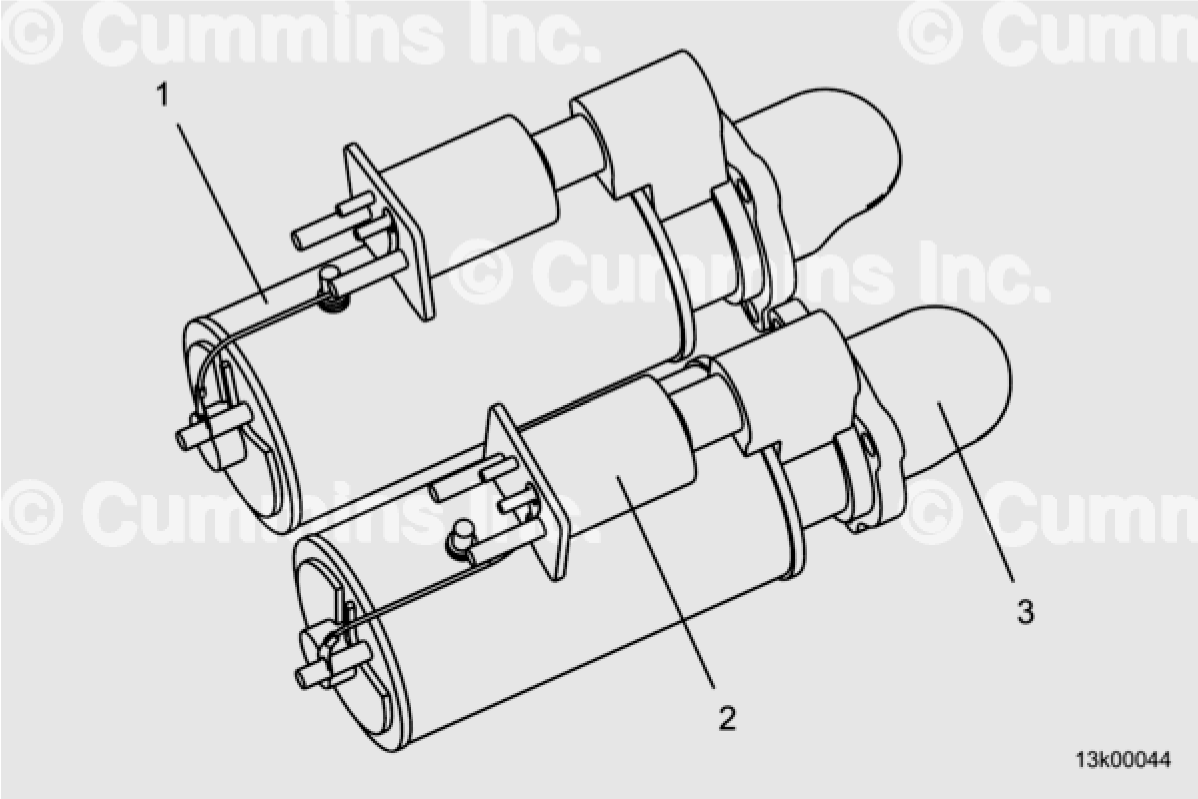
There are two starter motor options being offered:

- Delco Remy® Starting Motors
- Prestolite® Starting Motors



Prestolite® Starting Motor Exploded View

1. Starting motor
2. Starter solenoid
3. Starter magnetic switch
4. Pinion



Delco Remy® Starting Motor Exploded View

1. Starting motor
2. Starter solenoid
3. Pinion

General Information

The Prestolite® and Delco Remy® electric starting motors may be provided by Cummins Inc. or the equipment manufacturer. If provided by the equipment manufacturer, see equipment manufacturer service information. The electric starting motors operate on 24 VDC. Power to the starting motors is provided via the original equipment manufacturer (OEM) wiring harness. The starting motors are mounted on the flywheel housing. The starter magnetic switch is mounted to the starting motor and connected to the OEM wiring harness. All components associated with the Prestolite® starting motor weigh less than 23 kg [50 lb]. All components associated with the Delco Remy starter motor weigh less than 36 kg [79 lb].

Preparatory Steps

⚠ WARNING ⚠

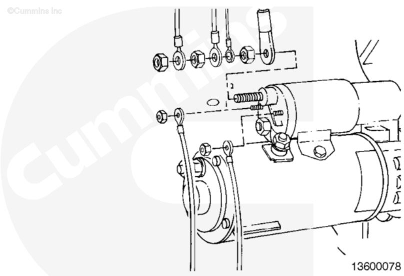
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the battery. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Disconnect the battery. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Remove the electrical connections from the starter motor.

Remove

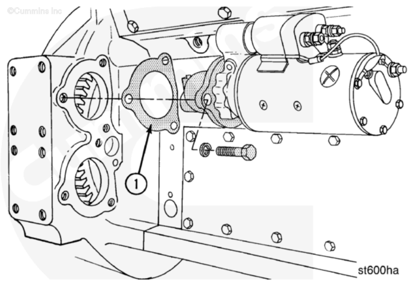
Delco Remy® Starting Motor

Remove the electrical connections from the Delco Remy® starting motor.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the starting motor capscrews, starter, spacers, and gaskets.

Discard the gaskets.

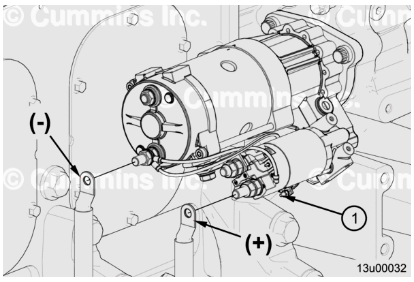
Note : Not all engines contain spacers and gaskets.

Prestolite® Starting Motor

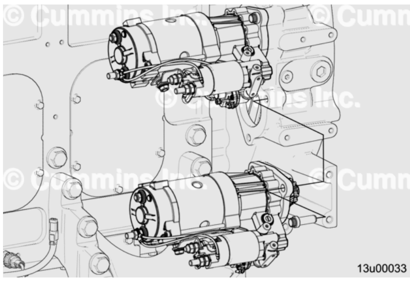
Remove signal wire from starter magnetic switch (1).

Remove positive (+) cable from starter solenoid.

Remove negative (-) cable from starting motor.



Remove starting motor.



Clean and Inspect for Reuse

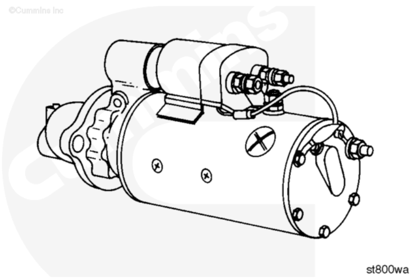
Delco Remy® Starting Motor

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use steam to clean the exterior of the starting motor.

Check the gear, shaft, terminal posts, and bushing for wear or damage.

Replace the starting motor if worn or damaged.

Use engine oil to lubricate the bushing.

A pipe plug **must** be removed to lubricate the bushing on some starter motors.

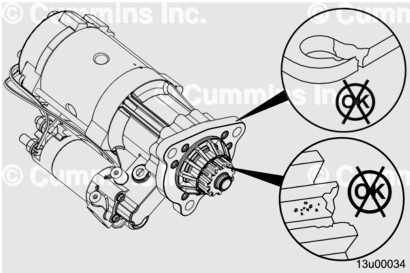
Prestolite® Starting Motor

Clean exterior of starting motor. Use steam. Dry with compressed air.

Inspect starting motor body and pinion.

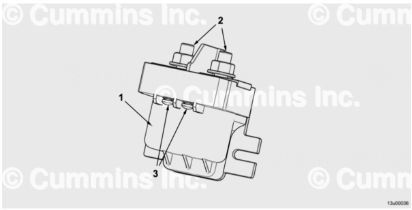
Replace starting motor if:

- Cracked
- Pinion teeth are worn
- Pinion teeth are chipped
- Pinion teeth are broken
- Otherwise damaged



Inspect starting motor wiring for corroded or loose connections.

Clean and tighten connections, if necessary.

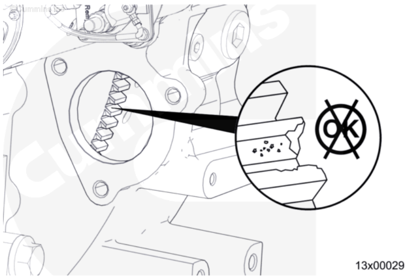


Use engine barring device to rotate engine one full turn while inspecting teeth of flywheel ring gear through starter flange hole.

Replace ring gear if:

- Worn
- Otherwise damaged

Refer to Procedure 016-008 in Section 16. (/qs3/pubsys2/xml/en/procedures/102/102-016-008.html)

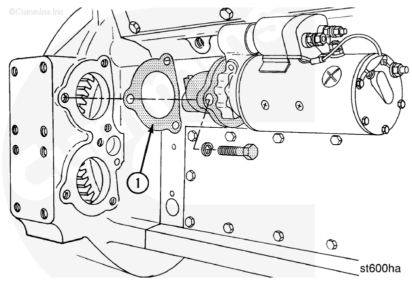


Install

Delco Remy® Starting Motor

⚠ CAUTION ⚠

Make sure to use the same thickness of starter motor spacer (1), (if used), as the one removed to install the starter motor to prevent engine or starter motor damage.



Installation of the top electric starting motor first on K38 and K50 series engines will simplify the installation of both starting motors.

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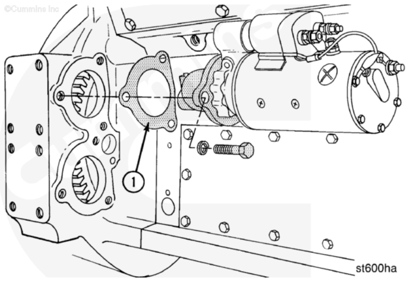


Note : Not all engines use spacers.

The wet type of flywheel housing requires gaskets for the starter motor.

Install any spacers or gaskets.

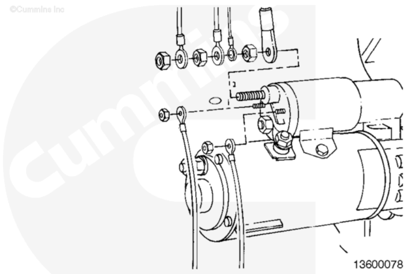
Install the starter motor and capscrews.

Tighten the capscrews.

Torque Value:
Cast Iron Flywheel Housing 215 n•m [159 ft-lb]

Torque Value:
Aluminum Flywheel Housing 195 n•m [144 ft-lb]

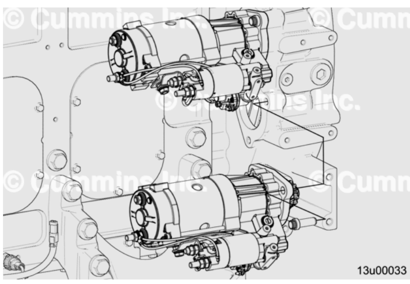
Install the electrical connections to the starting motor.



Prestolite® Starting Motor

Install starting motor.

Torque Value: 215 n•m [159 ft-lb]



Connect positive (+) cable to "B" terminal on starter solenoid.

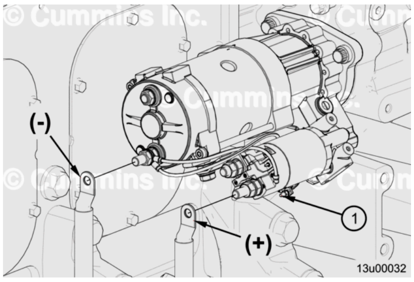
Torque Value: 26 n•m [230 in-lb]

Connect negative (-) cable to terminal on starting motor body.

Torque Value: 26 n•m [230 in-lb]

Connect signal wire to terminal on starter magnetic switch (1).

Torque Value: 2.25 n•m [20 in-lb]



Finishing Steps

⚠ **WARNING** ⚠

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- Connect the battery. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the starter and check for proper operation.

- Install the electrical connections to the starter motor.
- Connect the battery. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the starter and check for proper operation.